# Alexandria National University

## Faculty of Computer and Data Science

## Project Tittle: Game Theory_(01416N)

## Project Title: Strategic Competition Simulation


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# Strategic Competition Simulation Using Game Theory

## Cournot Competition Between Two Firms

## 1. Problem Description

This project studies strategic interaction between two firms competing in a market using Game Theory concepts.

The model is based on **Cournot Competition**, where each firm decides its production quantity independently while considering the competitor's decision.

The project aims to analyze:

- Strategic behavior of firms
- Nash Equilibrium
- Best response dynamics
- Market efficiency
- Impact of cost and demand changes
- Leader–Follower (Stackelberg) competition

The system is fully implemented using Python and includes simulation, visualization, and analytical solutions.

## 2. Game Modeling

## Players

- Firm A
- Firm B

## Strategy Space

The model uses:

**Quantity Competition (Cournot Model)**

Each firm chooses:

for Firm A $q_1$
for Firm B $q_2$

## Market Price Function

$$P = a - b(q_1 + q_2)$$

## Profit Functions

### Firm A Profit

$$\pi_1 = (P - c_1)q_1$$

### Firm B Profit

$$\pi_2 = (P - c_2)q_2$$

## 3. Theoretical Analysis

# Best Response Functions

### Firm A

$$BR_1(q_2) = \frac{a - c_1 - bq_2}{2b}$$

### Firm B

$$BR_2(q_1) = \frac{a - c_2 - bq_1}{2b}$$

# Nash Equilibrium

### Equilibrium Quantities

$$q_1^* = \frac{a - 2c_1 + c_2}{3b}$$

$$q_2^* = \frac{a - 2c_2 + c_1}{3b}$$

At equilibrium:

- Each firm is playing its best response
- No firm can improve profit unilaterally

# 4. Game Representation

## Normal Form Representation

The payoff matrix is implemented using Python dictionaries / arrays.

```
========== NORMAL FORM PAYOFF MATRIX ==========

Firm A Quantity = 10
Firm B Quantity = 10 --> Payoffs (600, 600)
Firm B Quantity = 20 --> Payoffs (500, 1000)
Firm B Quantity = 30 --> Payoffs (400, 1200)
Firm B Quantity = 40 --> Payoffs (300, 1200)

Firm A Quantity = 20
Firm B Quantity = 10 --> Payoffs (1000, 500)
Firm B Quantity = 20 --> Payoffs (800, 800)
Firm B Quantity = 30 --> Payoffs (600, 900)
Firm B Quantity = 40 --> Payoffs (400, 800)

Firm A Quantity = 30
Firm B Quantity = 10 --> Payoffs (1200, 400)
Firm B Quantity = 20 --> Payoffs (900, 600)
Firm B Quantity = 30 --> Payoffs (600, 600)
Firm B Quantity = 40 --> Payoffs (300, 400)

Firm A Quantity = 40
Firm B Quantity = 10 --> Payoffs (1200, 300)
Firm B Quantity = 20 --> Payoffs (800, 400)
Firm B Quantity = 30 --> Payoffs (400, 300)
Firm B Quantity = 40 --> Payoffs (0, 0)
```

## Extensive Form Representation

The game is represented as a decision tree:

```
========== EXTENSIVE FORM TREE ==========

Firm A chooses: Low Quantity
   -> Firm B responds: Low
   -> Firm B responds: Medium
   -> Firm B responds: High

Firm A chooses: Medium Quantity
   -> Firm B responds: Low
   -> Firm B responds: Medium
   -> Firm B responds: High

Firm A chooses: High Quantity
   -> Firm B responds: Low
   -> Firm B responds: Medium
   -> Firm B responds: High
```

This represents:

- Firm A moves first
- Firm B responds after observing A's decision

## 5. Python Implementation

The project is implemented using:

- Object-Oriented Programming (OOP)
- NumPy for computations
- Matplotlib for visualization

### Main Classes:

- `CournotCompetition`
- `ExtensiveFormGame`
- `SimulationManager`

The system supports:

- Parameter changes (cost, demand, sensitivity)
- Multiple simulations
- Scenario comparison

## 6. Simulation Scenarios

The model is tested under different market conditions:

# Scenario 1: Base Case

Standard market conditions.

**Base Nash Equilibrium Output**

```
========== NASH EQUILIBRIUM ==========
Firm A Quantity : 26.67
Firm B Quantity : 26.67
Market Price : 46.67
Firm A Profit : 711.11
Firm B Profit : 711.11
```

# Scenario 2: High Cost (Firm A)

Firm A has higher production cost.

**Cost Impact**

```
Scenario: High Cost for Firm A
====================================
Firm A Quantity : 16.67
Firm B Quantity : 31.67
Market Price : 51.67
Firm A Profit : 277.78
Firm B Profit : 1002.78
```

# Scenario 3: Demand Increase

Market demand increases.

```
Scenario: Demand Increase
==========================
Firm A Quantity : 40.00
Firm B Quantity : 40.00
Market Price : 60.00
Firm A Profit : 1600.00
Firm B Profit : 1600.00
```

# Scenario 4: Demand Decrease

Market demand decreases.

```
Scenario: Demand Decrease
===========================
Firm A Quantity : 16.67
Firm B Quantity : 16.67
Market Price : 36.67
Firm A Profit : 277.78
Firm B Profit : 277.78
```

## Scenario 5: High Market Price Sensitivity

Price becomes more sensitive to quantity changes.

```
Scenario: High Market Price Sensitivity
====================================
Firm A Quantity : 66.67
Firm B Quantity : 66.67
Market Price : 53.33
Firm A Profit : 2222.22
Firm B Profit : 2222.22
```

## Stackelberg Competition (Leader–Follower)

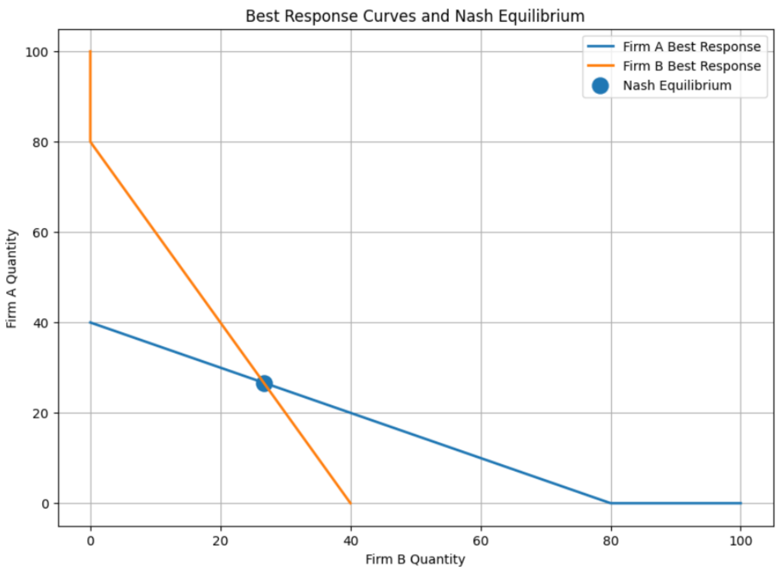
Firm A acts as leader, Firm B follows.

```
========== STACKELBERG EQUILIBRIUM ==========
Leader Quantity : 40.00
Follower Quantity : 20.00
Market Price : 40.00
Leader Profit : 800.00
Follower Profit : 400.00
```

## 7. Visualization Results

The project generates multiple visual outputs:

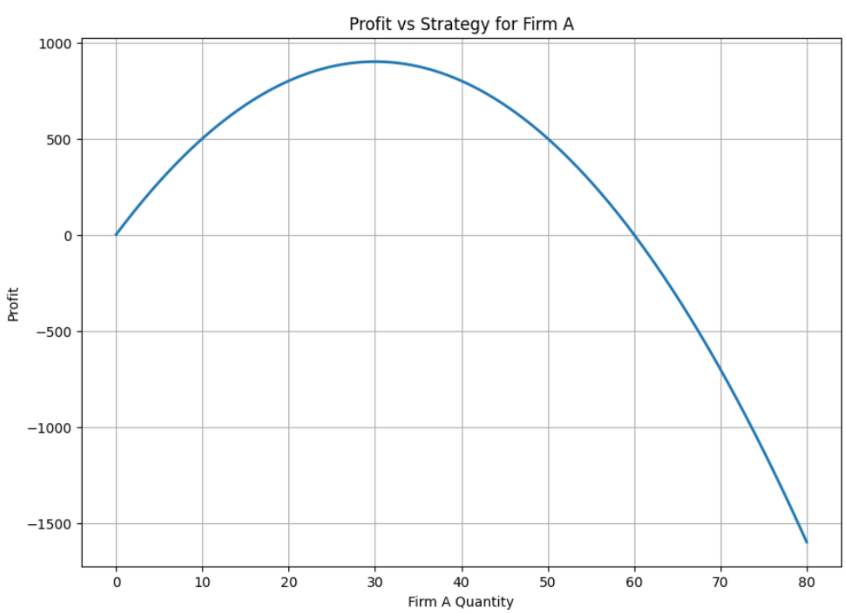
## Best Response Curves




Shows:

- Intersection point = Nash Equilibrium

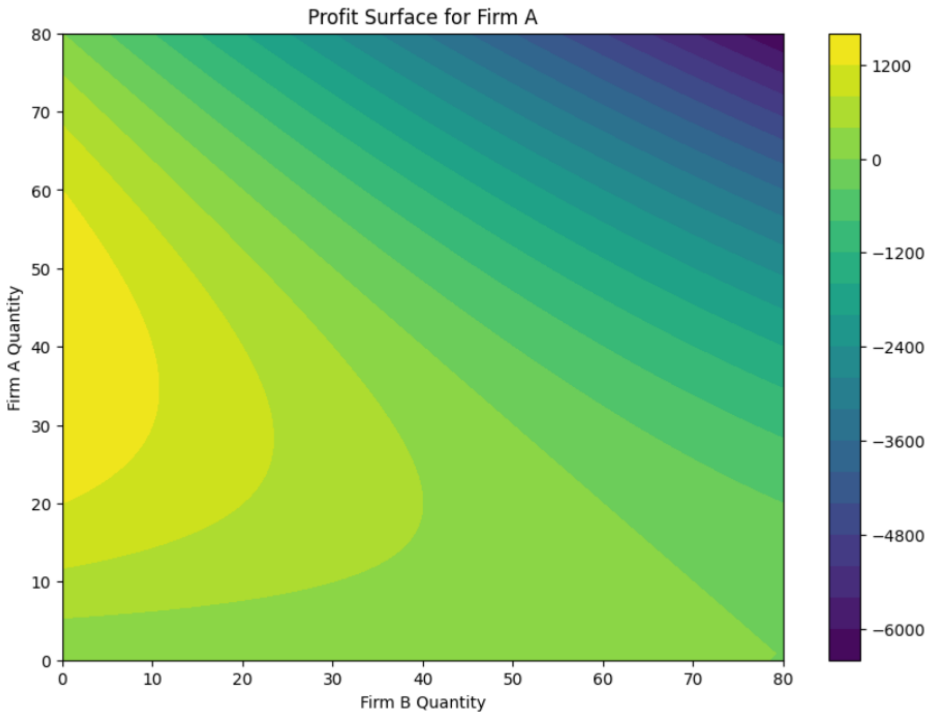
## Profit vs Strategy




Shows how profit changes with quantity.

## Profit Surface

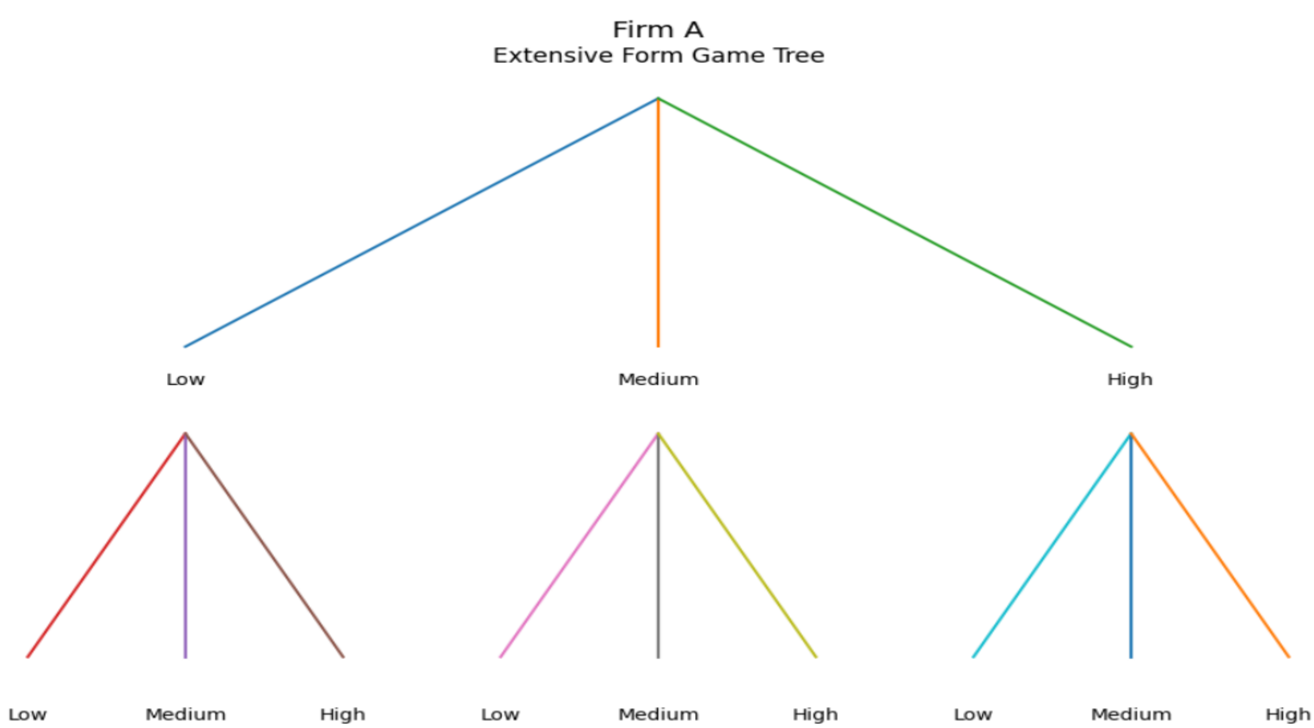
### 3D/Contour Profit Surface




Shows profit distribution across strategies.

## Extensive Form Tree

### Game Tree Visualization



## 8. Results

The simulation shows:

- Firms always adjust strategies based on competitors
- Nash equilibrium is stable across scenarios
- Market demand strongly affects output and profit
- Higher costs reduce competitiveness
- Strategic interdependence is clearly visible

## 9. Insights

## Why Equilibrium Occurs

Equilibrium occurs because:

- Each firm selects its best response
- No unilateral deviation improves payoff

## Efficiency

Cournot equilibrium is:

- Stable
- But not socially optimal

Because:

- Firms restrict output
- Prices remain above marginal cost

## Real-World Applications

This model applies to:

- Airlines
- Oil industry
- Telecom markets
- Tech competition (Intel vs AMD, Boeing vs Airbus)

## 10. Conclusion

This project successfully models strategic market competition using Game Theory.

It includes:

- Mathematical modeling
- Nash equilibrium analysis
- Simulation under multiple scenarios
- Visualization of strategic behavior
- Extensive and normal form games
- Stackelberg competition

The results demonstrate how firms behave strategically in real-world competitive markets.

### ***Scan Qr_Code To get Source Code :***